1st Semester 2024 / 2025

FINAL EXAM

Time Allowed: 3 Hours

Course	:	QM501E_MSC - Business Analytics
Instructors	:	Dr. Ahmed ATIL

STUDENTS INSTRUCTIONS	
ANSWER	Answers must be in English.
DICTIONARY	Only RSB English dictionary allowed. It will be available upon request to the invigilator
DOCUMENTATION	Documentation allowed. Computer allowed: Students are allowed to use their own documentation as well as their laptop computers and all the information contained. Students are not allowed to share notes.
APPENDIX	Dominos_Pizza_Sales.xlsx

ILOS TESTED BY THE EXAM:

1- Describe the major components of business analytics
2- Understand how to explore and visualize data using MS Excel & Python
3- Apply basic statistical modelling and sampling using MS Excel & Python
4- Understand how to extract useful information from data (Python)
5- Understand the purpose of data mining and machine learning (Python)

100 MARKS IN TOTAL, AWARDED ACCORDING TO THE FOLLOWING TABLE:

Question 1	50 points
Question 2	30 points
Question 3	20 points
Total	***100 points***

FINAL EXAM
QM501E_MSC - Business Analytics

During the exam, it is strictly forbidden to:

- ✓ **Use the mobile phone network (4G, 5G).**
- ✓ **Activate any Virtual Private Network (VPN)**
- ✓ **Use private or incognito browsing modes**
- ✓ **Connect to internet web sites or applications, including those related to OpenAI**

Furthermore, please note that when you click to start the exam, **you are providing legal consent to be monitored** for the entire duration of the examination.

Failure to comply with any of these rules may result in **severe consequences**, ranging from **receiving a grade of zero** for the exam to facing **disciplinary measures**

Data Analytics internship at Domino's Pizza

Project Context:

You are a master's student completing an internship at Domino's Pizza, one of the largest pizza chains in the world, specializing in fast pizza delivery. Founded in 1960, the company operates in more than 90 countries with thousands of locations. Domino's Pizza not only focuses on delivering pizzas but also leverages data to improve its operations, optimize team performance, and enhance customer satisfaction.

During this internship, you are integrated into the **Data Analytics team**, led by an experienced **Data Scientist**, who collaborates closely with the **Sales Manager** of the company. Your mission is to work on a key project that has been identified as a priority by the sales team.

Scenario:

The sales manager, **Léa Morel**, enthusiastically welcomes you during your first meeting and starts by explaining the current challenges Domino's Pizza is facing.

"Our goal is to increase our teams' performance and deeply understand our customers' consumption habits. We have a wealth of information thanks to our databases, but to fully exploit it, we need an expert eye and advanced analytical techniques. You'll be working with our Data Scientist, Mr Atil, who will guide you through the process."

You quickly realize that your task will involve not only cleaning the data but also extracting actionable **insights**. Ahmed, your mentor for this internship, explains the key steps of your mission.

"We have a lot of data: orders, customer satisfaction, seller performance… Your job will be to perform Exploratory Data Analysis (EDA) to understand what the numbers are telling us. Once the data is cleaned and structured, we'll move on to predictive modeling. Our goal is not only to analyze the past but also to predict customer satisfaction going forward. That's what Léa is most interested in."

Mr. Atil reassures you that you won't be working alone, but he expects you to work independently on well-defined tasks. You'll also attend meetings with Léa and the sales team to present your findings, which will be used to make strategic decisions.

Project Objectives:

Léa Morel outlines the three main objectives for this project:

1. **Analyze Sales and Seller Performance**
   Léa wants to understand which pizzas are selling the best and which sellers are performing the most. Analyzing the product segments (types of pizzas) and employee performance will be central to this objective.

   > *"I need you to identify which pizzas are generating the most revenue. At the same time, I want a ranking of the top sellers based on their sales in euros. We need to know who is making a difference on the field."*

2. **Visualize Sales and Seller Performance**
   To make it easier to communicate the results to the team, Léa emphasizes the importance of visualization.

   > *"I want clear and precise charts that show seller performance and product segments. Numbers are good, but visuals speak better!"*

   Your task will be to use visualization tools to transform the data into charts and dashboards that provide a simple and effective overview of performance.

3. **Predict Customer Satisfaction**
   The final objective is future-oriented. Ahmed explains what you'll need to do to help Léa better understand customer satisfaction.

"We'll need to predict customer satisfaction based on certain criteria. We'll use a machine learning model to classify customers based on their satisfaction level. This will allow us to anticipate and proactively improve the customer experience."

Customer satisfaction is measured by a satisfaction rate, which you will transform into three classes (low, medium, and high satisfaction). You will then build a predictive model to estimate customer satisfaction based on the available variables (e.g., pizza type, seller performance, etc.).

Instructions:

- Use the dataset " **Dominos_Pizza_Sales.xlsx "** provided by Domino's Pizza
- Perform all tasks using Python within a **Jupyter Notebook.**
- Justify all your model choices and interpretations with **appropriate statistical reasoning.**
- All **comments, remarks, and conclusions must be written** in the **same Jupyter notebook** used to produce your code. This ensures that the analysis is well-documented, transparent, and easy to follow for anyone reviewing the project.

Project Steps:

Question 01 - Step 1: Descriptive Data Analysis (EDA) (50 points)

1.1 Data Preprocessing

Your first task will be to clean the data to extract useful insights. Ahmed assigns you the following steps:

- **Remove Duplicate Rows**: Identify and remove duplicates from key columns (order_details_id, order_id) to avoid skewing the analysis.
- **Remove Irrelevant Values**: Clean inconsistent or extreme values in columns like unit_price, quantity, and total_price.
- **Handle Missing Values**: Use dropna() or fillna() to handle missing data, explaining the strategy you choose.
- **Encode Categorical Data**: Convert categorical data (pizza_category, GenderOfTipper) into numeric formats using methods like One-Hot Encoding or Label Encoding.

1.2 Sales Analysis by Pizza Segment

You will analyze the sales by segmenting products (by pizza category and type) to identify which pizzas generate the most revenue.

1.3 Seller Performance

Identify the top seller based on total sales revenue. This analysis will provide Léa with a ranking of the best-performing employees.

1.4 Visualization of Results

Using tools like Excel or Python (with libraries such as Matplotlib or Seaborn), you will create visualizations to showcase:

- The most popular pizza segments.
- Revenues generated by seller & pizza Type.

Question 02 - Step 2: Predictive Analytics (30 points)

2.1 Transforming the Satisfaction Variable
You will transform the Satisfaction_Rate column into a new variable "Satisf-Class" with three classes:

- **Class 1**: Satisfaction rate < 0.33
- **Class 2**: 0.33 ≤ Satisfaction rate ≤ 0.66
- **Class 3**: Satisfaction rate > 0.66

2.2 Predictive Modelling You will build a machine learning model (logistic regression, Random forest , or decision tree) to predict customer satisfaction based on available variables. Your tasks include:

- Training the model on part of the dataset.
- Testing the model to evaluate its performance (precision, recall, F1-score).
- Interpreting the results and presenting the key factors influencing customer satisfaction.

Question 03 - Step 3: Final Report Analytics (20 points)

Using the Jupyter notebook with plain text options, produce a detailed synthesis report in the form of a final conclusion for each phase of the project:

For **Step 1 (Descriptive Analytics)**, you should summarize the key insights derived from your data analysis, focusing on sales performance, product segmentation, and seller performance. Ensure that you present clear conclusions for each objective and provide justifications based on the data analysis, visualizations, and metrics you have calculated.

For **Step 2 (Predictive Analytics)**, present a comprehensive conclusion regarding the model's ability to predict customer satisfaction. Discuss the accuracy of your model, the key factors influencing customer satisfaction, and the practical implications of your results. Justify your conclusions by referring to model performance metrics, feature importance, and the classification results of the 'Satisf-Class' variable.

In both steps, ensure that your conclusions are well-justified and supported by the data and analysis results.

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